CASE STUDY

Scaling Inventory Without Scaling Headcount

in Automotive Manufacturing

Tier-1 Supplier | 4 Plants | 3 Warehouses | 5,000+ SKUs

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4

Plants

3

Warehouses

5,000+

SKUs

Business Challenge

Manual Tracking

Spreadsheets and legacy systems for inventory tracking across 4 plants

Manpower Dependency

Increasing headcount for every rise in stock handling volume

Stock Mismatches

Frequent discrepancies and delayed reconciliation causing confusion

Poor Visibility

No unified view of RM, WIP, and FG across locations and warehouses

Inefficient Movement

Material transfers between plants and warehouses poorly managed

Line Stoppages

High risk of production halts due to material unavailability

Solution Implemented

Smart Inventory Management System — Integrated with ERP & MES

Real-Time Tracking

RM, WIP & FG visibility across all locations simultaneously

Barcode / QR Handling

Fast and error-free stock transactions via scanning

Automated Movements

System-driven transfers between stores and production

Central Dashboard

Live stock levels, shortage alerts and excess tracking

Auto Reorder Controls

Min-max logic to prevent stockouts and overstocking

Bin Management

Optimized storage, location tracking and retrieval

ERP / MES Integration

Real-time material requirement alignment with production

Mobile Transactions

Enabled mobile-based stock operations for floor teams

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Implementation Approach

Phase 01

Process Mapping

- Analyzed existing inventory workflows
- Identified manual intervention points

Phase 02

Digitization

- Barcode system across warehouses
- Standardized SKU and bin structures

Phase 03

System Integration

- Connected with production modules
- Linked to procurement workflows

Phase 04

Training & Adoption

- Trained store operators & supervisors
- Enabled mobile-based transactions

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Results & Impact

40%

Efficiency Gain

inventory handling

60%

Less Manual Work

data entry reduced

2X

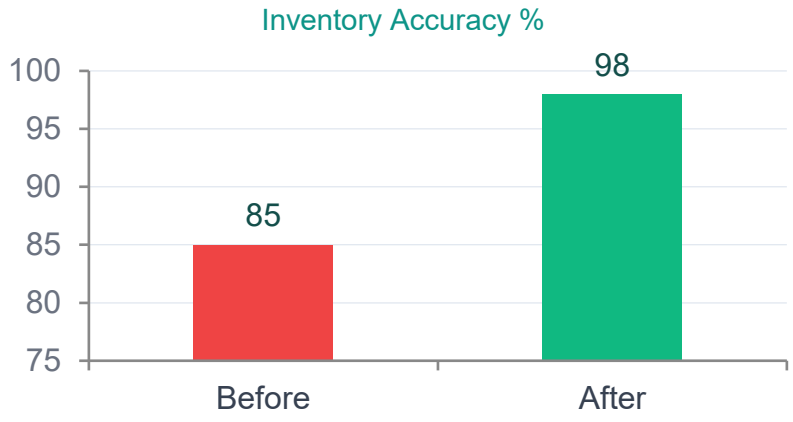
Volume Growth

no new headcount

25%

Cost Savings

labor cost reduction



Additional Wins

- ✅ 30% fewer line stoppages
- ✅ Scalable ops without more staff
- ✅ Real-time visibility all locations
- ✅ Faster audits & reconciliation
- ✅ Improved on-time dispatch

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Before vs. After Snapshot

Metric	Before	After
Inventory Volume	Limited by manpower	**2X Increase**
Headcount	Increasing	**Stable (No Additions)**
Inventory Accuracy	85%	**98%**
Manual Work	High	**Minimal (60% Cut)**
Line Stoppages	Frequent	**Reduced by 30%**
Operational Costs	Rising Rapidly	**25% Reduction**

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Conclusion

Scale Operations. Not Headcount.

By implementing a digitized and automated inventory management system, the organization successfully scaled its operations without increasing headcount. The shift to real-time visibility and process automation enabled higher efficiency, lower costs, and a strong foundation for future growth in the automotive supply chain.

Volume: +2X	**Accuracy: 98%**	**Costs: -25%**	**Stoppages: -30%**